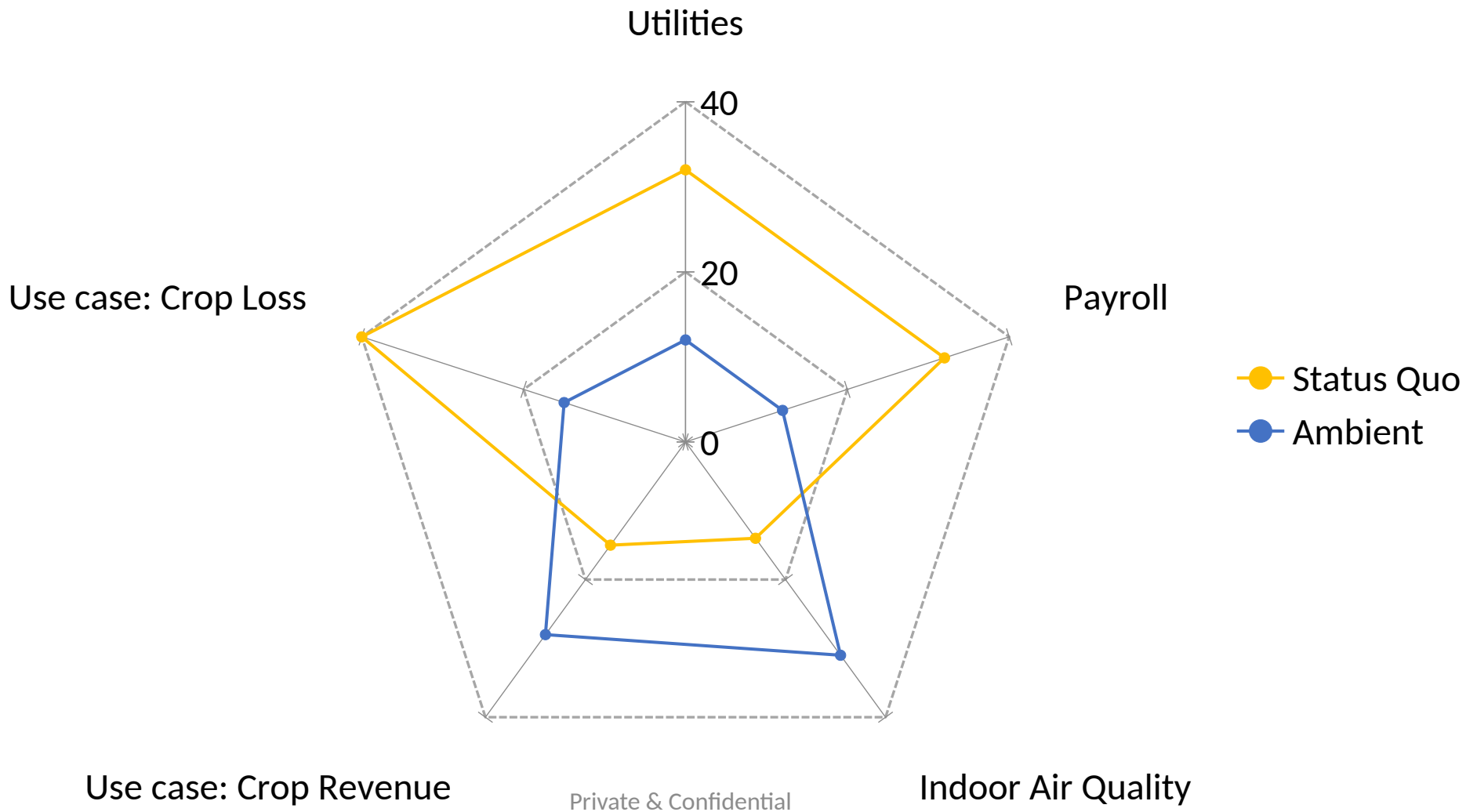
Ambient Systems

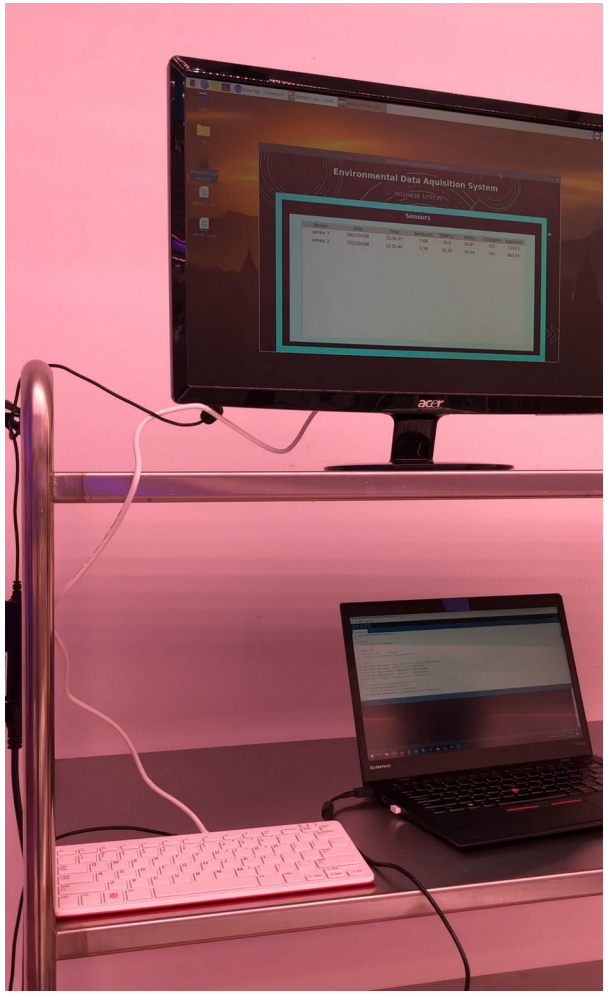
Business Plan Technology Slides

Problem: Multi-objective Optimization



Proof-of-Concept MVP Pilot

1. Collected climate data successfully
2. Identified gap between current and optimal
3. ine
4. Provided recommendations



Solution Targets Quantified

	Parameter	Actual	Optimal	Sample Action
Current POC	Air Temperature	28 C / 20 C	25 C / 12 C	Decrease temperature setpoint Increase ventilation
	Relative Humidity	55% to 70%	60% to 75%	Install fogging machines
	CO2	300 to 700 ppm	500 to 1200 ppm (min >330 ppm)	Decrease/increase ventilation (depends on CO2 concentrations outside) Install CO2 regulator
	Spectrum Light	2300 lux	5000+ lux	Turn off non-farming lights Reduce light intensity

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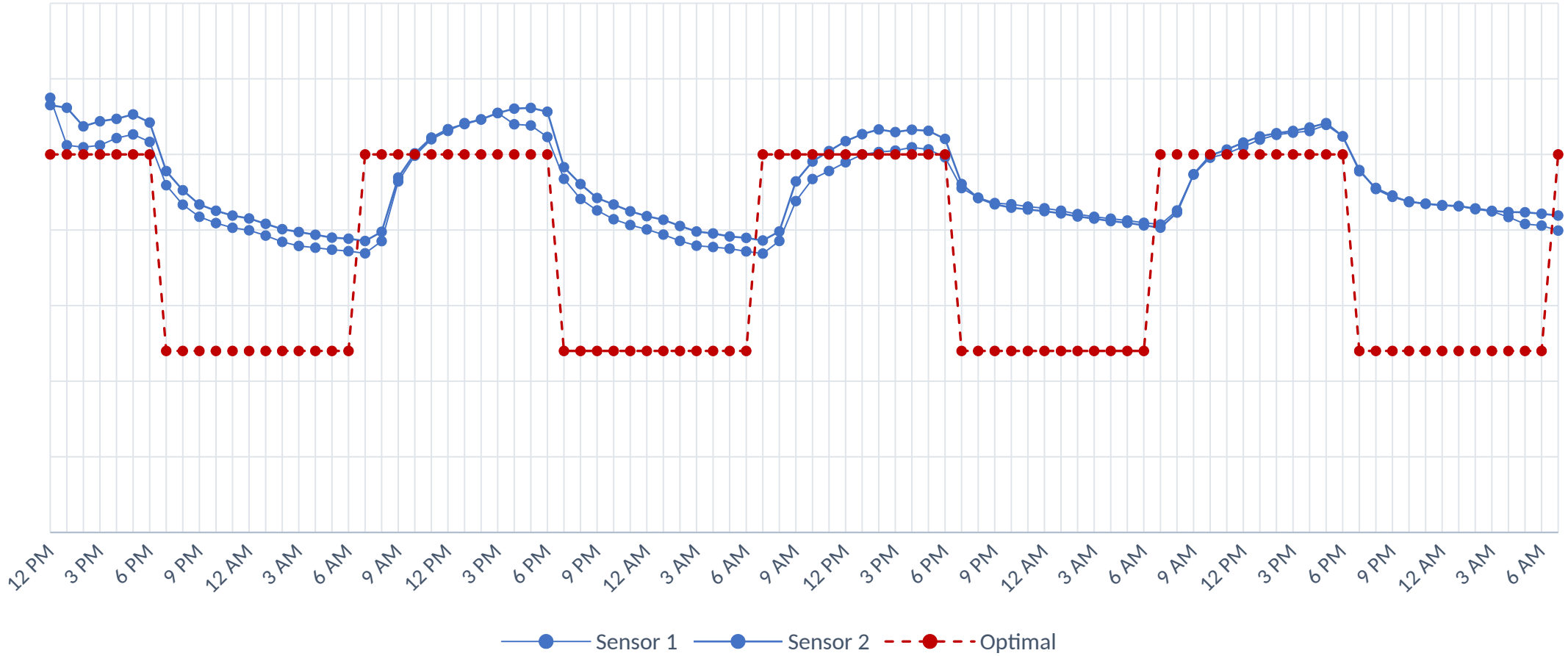
Full-scale Plot	Soil Temperature	MVP 2.0		
	Soil Moisture			
	pH level			
	Electrical conductivity (EC_w)			
	Crop specific parameters			

Temperature

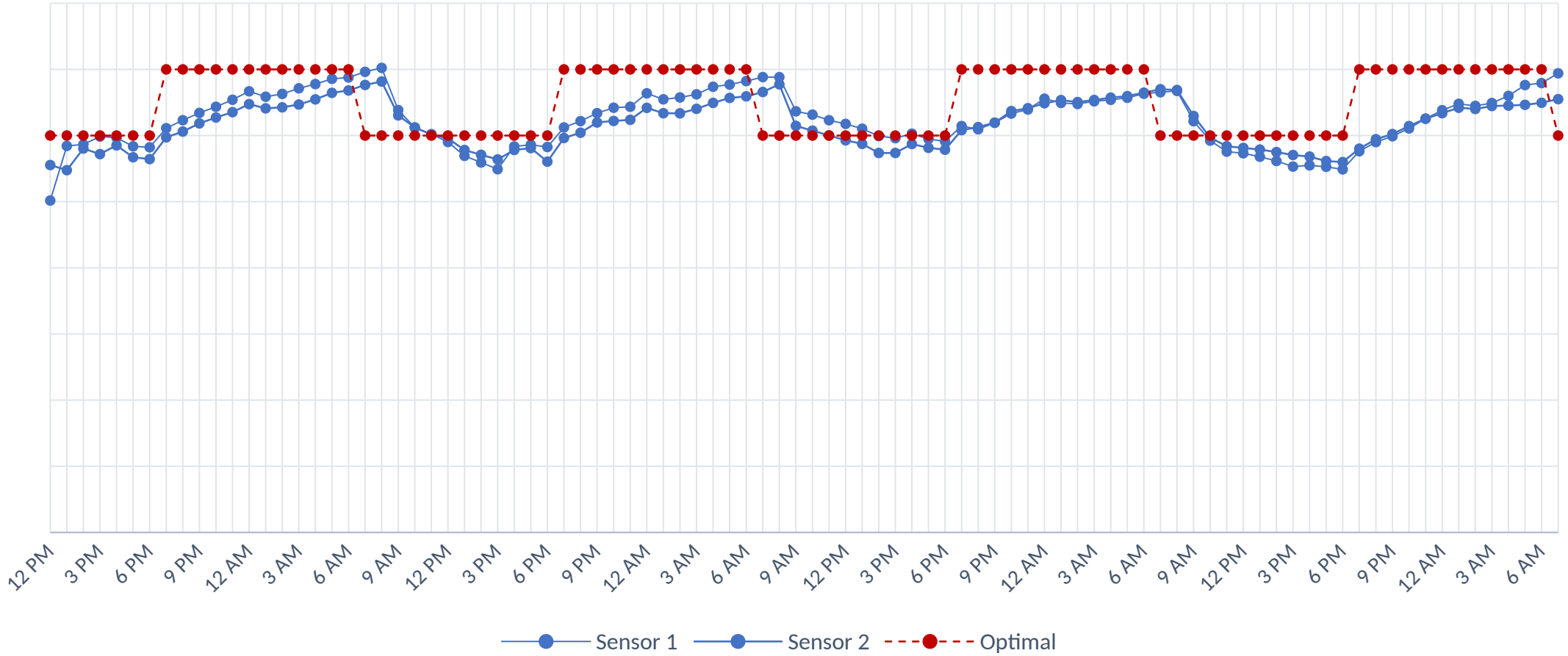
Leaf and petiole growth
Brighter leaf color
Greater sucrose
More biomass at the roots



Temperatures after bloom affect plant growth and fruit quality of strawberry, , 2000

Relative Humidity

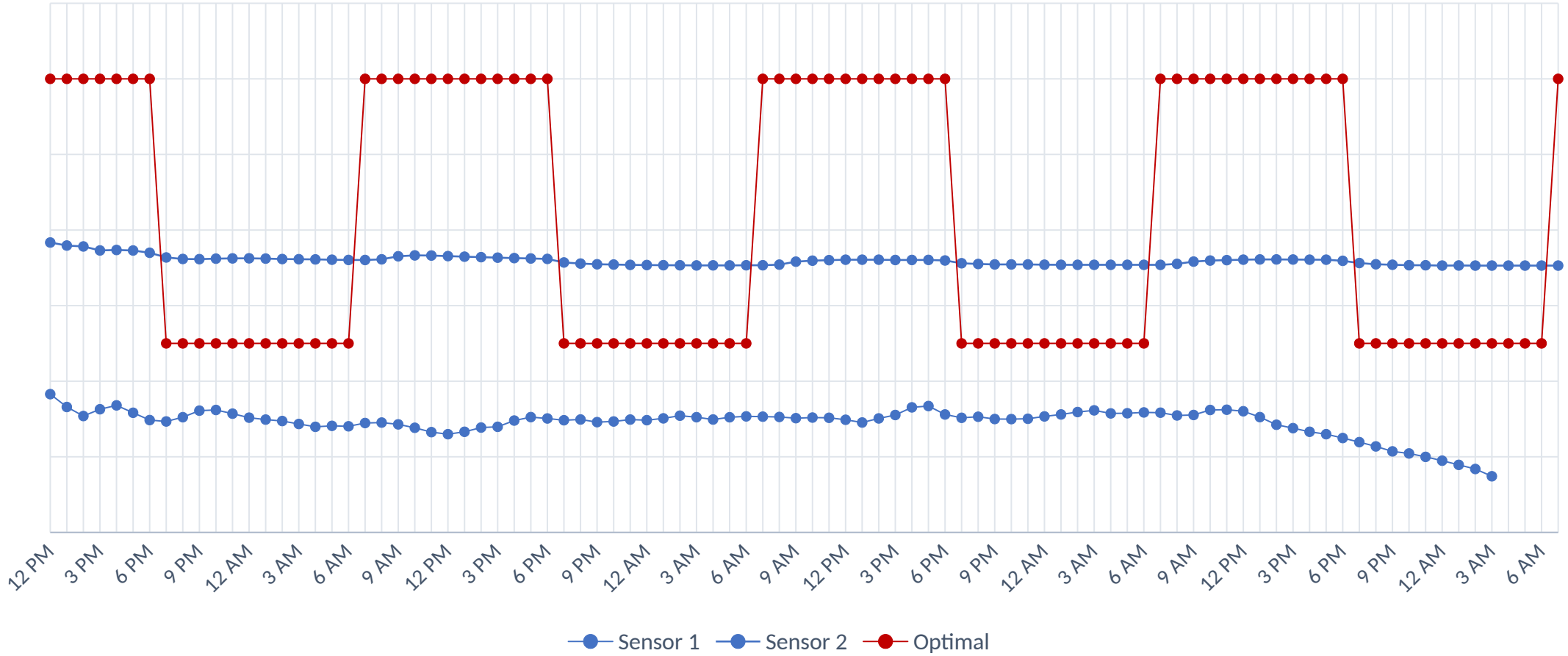
Avoid tip-burn and calyx burn
Avoid fungi
Avoid powdery mildew
Avoid leaf spot (bacterial blight)



Crop Guide: Growing Strawberry, , 2004

CO2

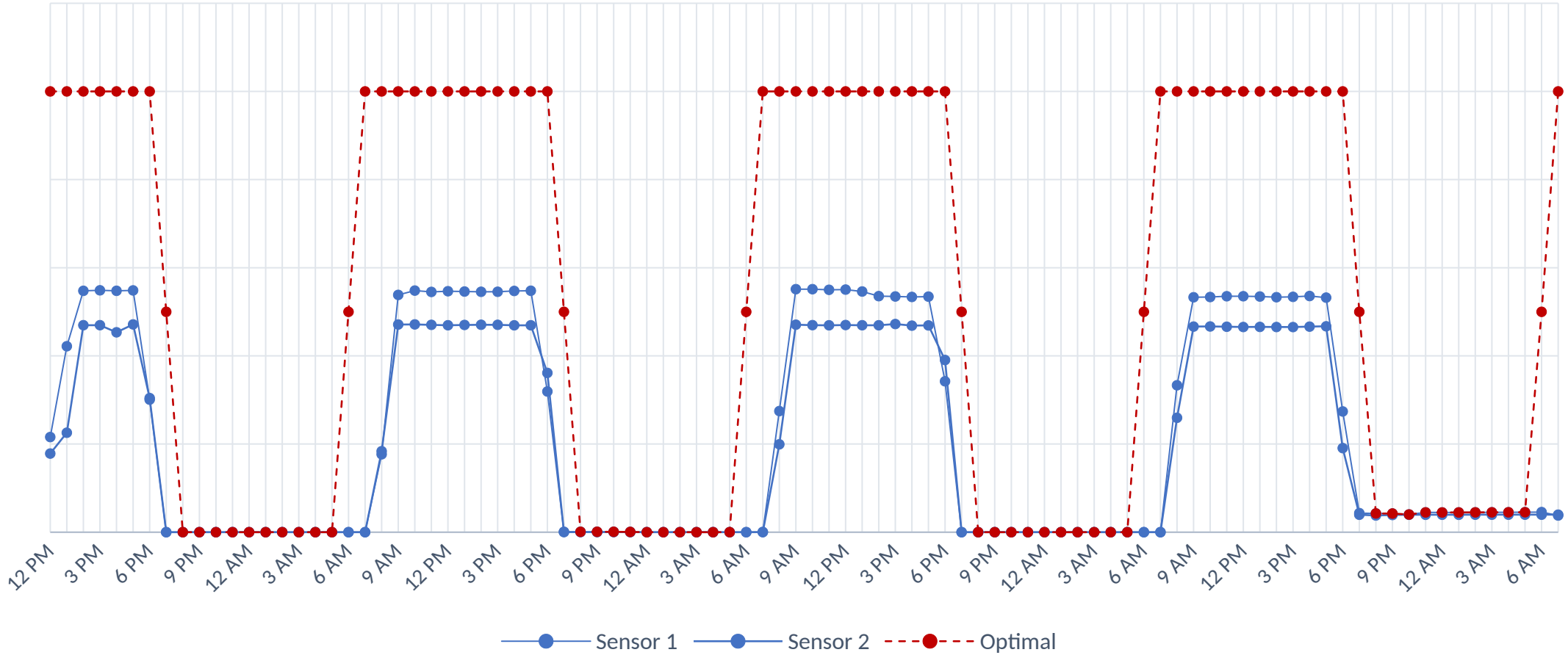
Increased photosynthesis
Earlier flowering
Higher fruit yields
Improved stem strength and flower size



Carbon Dioxide In Greenhouses, Ministry of Agriculture, Food and Rural Affairs, 2021
Carbon dioxide (CO2) applications in indoor growing,

Light Intensity

Increased photosynthesis

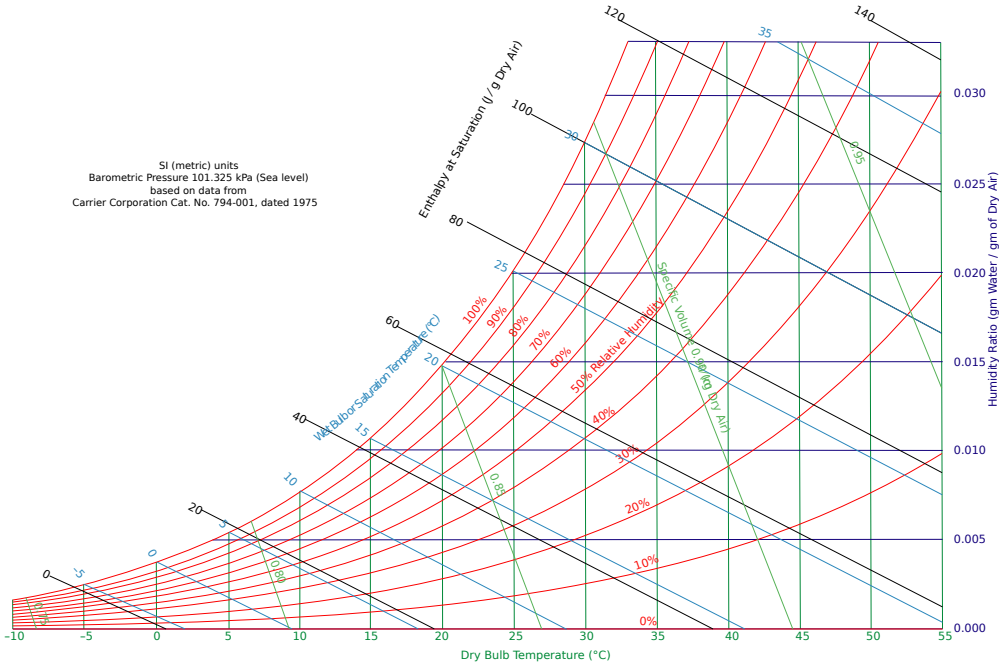


Effect Of Light Intensity, Temperature And Co2 Concentration On Photosynthesis And Growth Of In Vitro, International Society for Horticultural Science, 1989

Why not just change the AC settings?

#1: Interactions

- Psychometrics: temperature and relative humidity
- Light increases temperature and decreases humidity
- Decreasing temperature decreases CO2 concentrations in the air
- Increased ventilation introduces hot humid air f outside that needs to be cooled, purified, e hed, and re-humidified → very energy i sive → very expensive
- Safety: CO2 is needed by the plants, but human workers need oxygen to work indoors
- Photosynthesis process: lights and CO2 interact (both need to be high or both need to be low)
- Infiltration/exfiltration: porous building materials



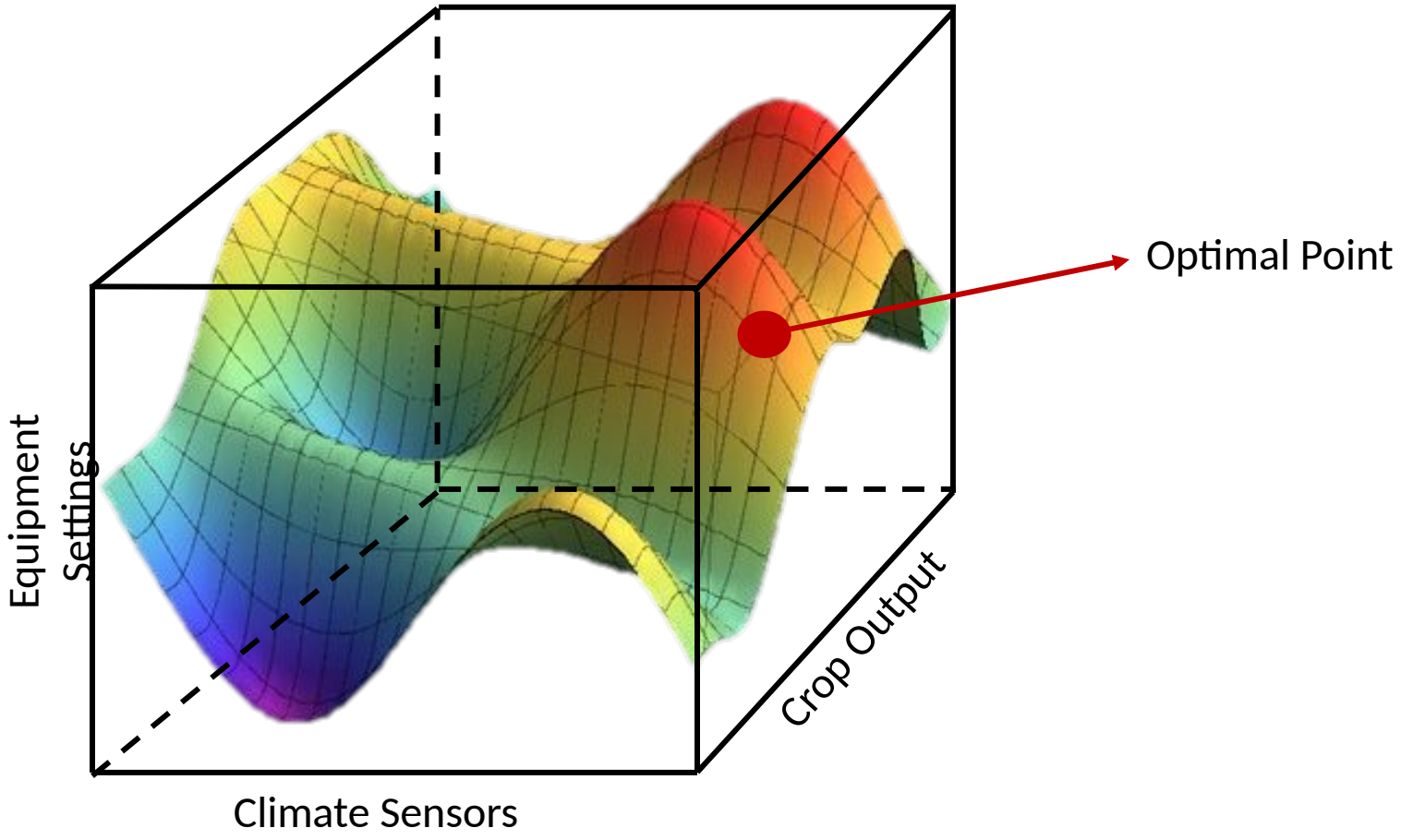
#2: Equipment

- Building equipment is often outdated
- AC controls are limited: only allows setting temperature and fan speed (not granular enough)
- AC controls cannot be managed remotely
- Need additional or custom equipment based on plant type, growing stage, and outdoor climate

How to address this?

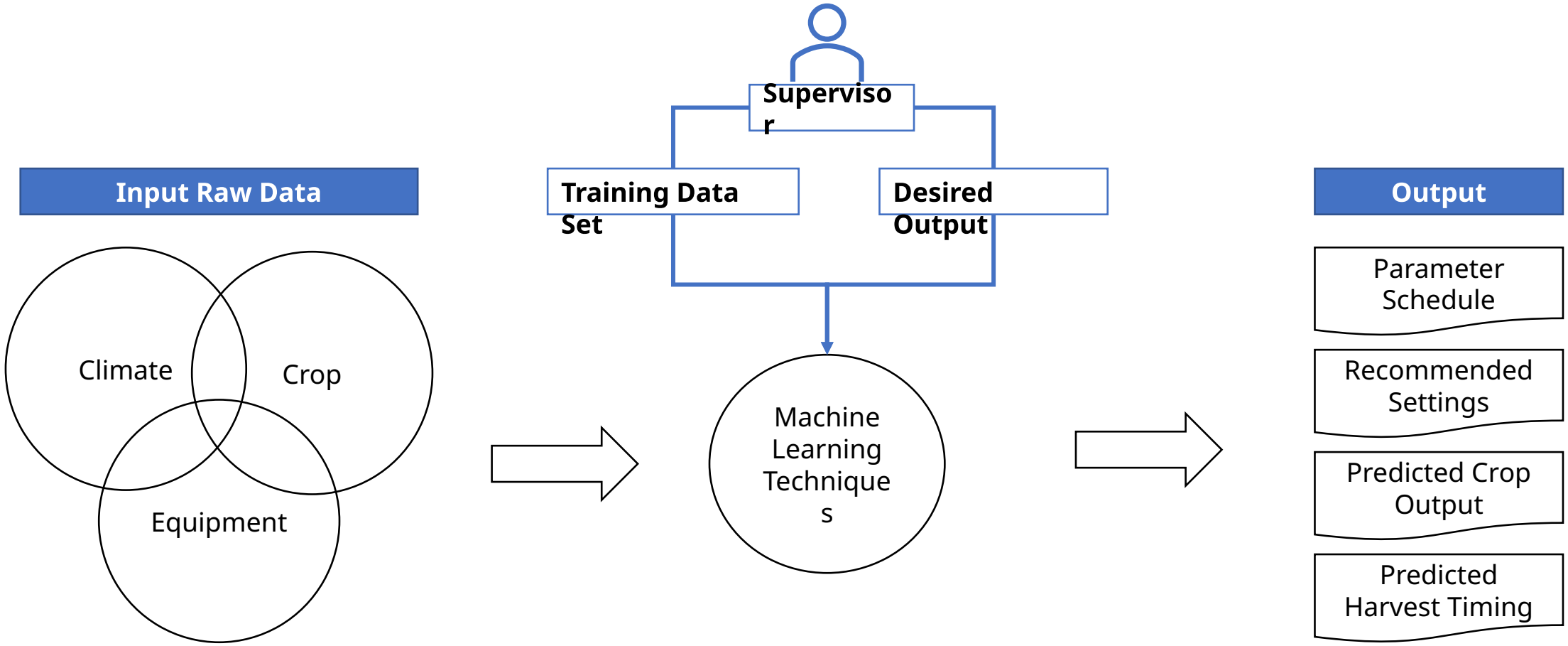
Optimization using building science-based AI models

Optimization: Multi-Dimensional Gradient Descent

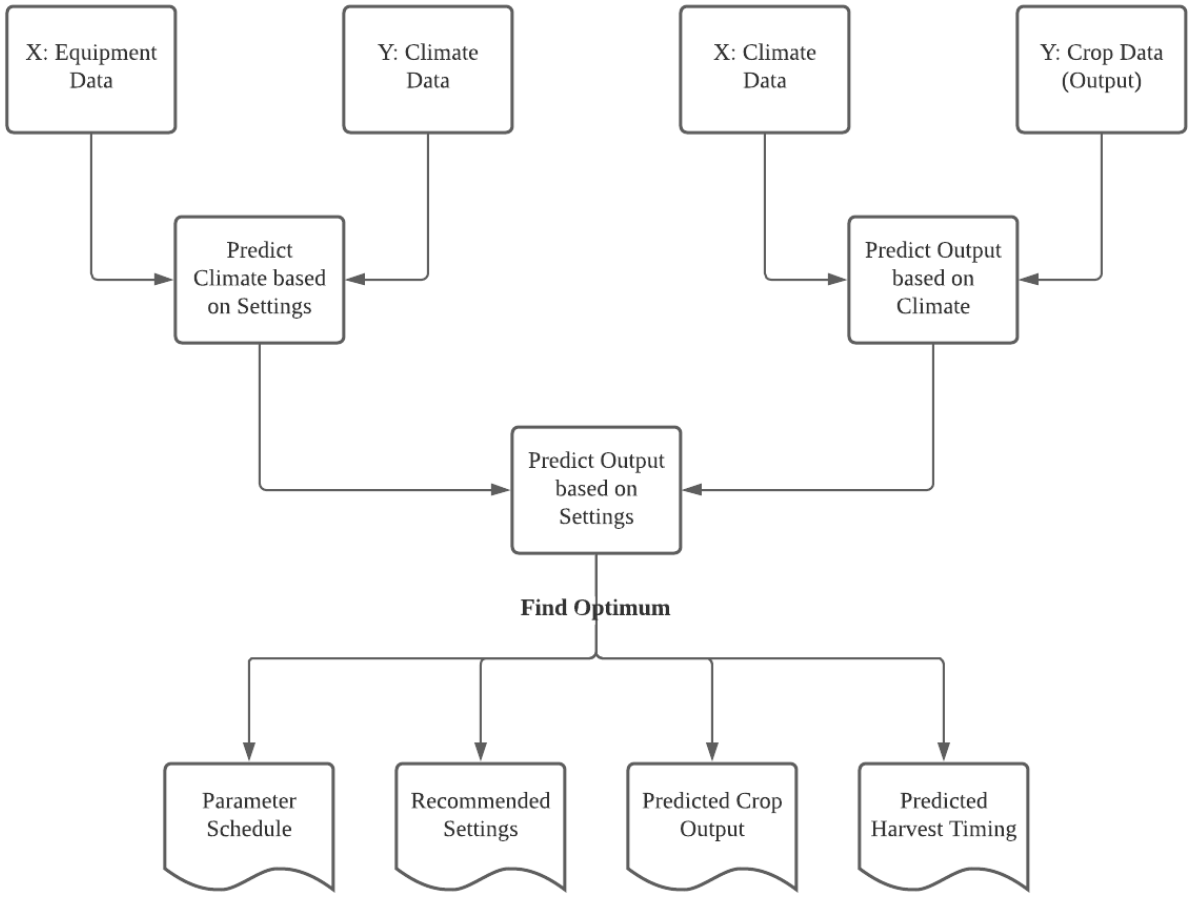


Ambient's Machine Learning Optimization Model

Patent



Supervised Learning: Three Stages



Notes:

- Non-climate parameters are already automated and controlled via dedicated systems (i.e., LED lighting, nutrition, irrigation, etc.)
- How Crop Data (output) is measured will vary based on farm type → it will impact data collected
- Product is agnostic to how crop output is defined or measured (assuming climate is an important input)
- More frequent data collection will increase the training of the feedback loop (e.g., providing crop data daily will allow quicker calibration than providing data at the end of each harvest)
- We can work with the farmer to provide the necessary data acquisition for crop quality (e.g., image recognition) and platform for training (e.g., optimization features)

Patent Classification

A	HUMAN NECESSITIES
B	PERFORMING OPERATIONS; TRANSPORTING
C	CHEMISTRY; METALLURGY
D	TEXTILES; PAPER
E	FIXED CONSTRUCTIONS
F	MECHANICAL ENGINEERING; LIGHTING; HEATING; WEAPONS; BLASTING
G	**PHYSICS** **G06:** COMPUTING; CALCULATING OR COUNTING **G06N:** COMPUTER SYSTEMS BASED ON SPECIFIC COMPUTATIONAL MODELS **G06N 3/00**: Computer systems based on biological models **G06N 5/00**: Computer systems using knowledge-based models **G06N 7/00**: Computer systems based on specific mathematical models **G06N 20/00**: Machine learning
H	ELECTRICITY

What are the benefits?

Productivity & Operational Efficiency

Ambient's optimization models show you how to create an optimal environment with:

- No need to manage sensors or raw data
- No need to manually adjust ACMV equipment early in the morning and late in the afternoon to accommodate plant growing schedule
- No need to manually adjust built-in (non-farming) lights
- No need for analysis of complex interactions between parameters
- No need to wait until seeing harvest results to fine-tune the parameters
- No need to deal with complex thermal inertia calculations
- No need for complex set-ups to perform farming experiments
- No need to spend time evaluating which mechanical equipment to buy (we recommend it)
- No need to worry about finding and selecting inter-operable systems (we shortlist them for you)

How much time can Ambient save you?

POC Pilot Outcomes

Recommendations

Insight	Recommendation	Cost
No historical data	Measure and securely archive critical operational data for crop repeatability and insurance	Ambient Platform Tier 1
Variability between days	Monitor for longer period of time (entire harvest cycle)	Ambient Platform Tier 1
No utilities data	Monitor energy consumption	$5k-$10k hardware modification Ambient Platform Tier 1 & 2
High temperature	Decrease daytime and night-time temperature	$5k - $10k portable AC
Low CO2	CO2 regulator (increase CO2)	$1k - $2k CO2 regulator $500 monthly CO2 supply
Moisture timing	Introduce mist earlier in the day Reduce mist later in the day	Ambient Platform Tier 1
Plants have various optimal parameter schedules	Determine farm portfolio optimal (not just for one plant but entire farm)	Ambient Platform Tier 1 & 3